

Introduction to Data Science and Engineering

- Web scraping



Some materials courtesy of
Rafael A. Irizarry, and are modified
from the original version.

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Some slides retrofitted by Michael Wu

- The data we need is not always available in a spreadsheet/table form. For example, the US murders dataset we used in the R Basics chapter originally comes from this Wikipedia page:

Gun violence in the United States by state

🌐 1 language ▾

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From Wikipedia, the free encyclopedia

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This article is a list of the [U.S. states](#) and the [District of Columbia](#), with [population](#), [murders](#) and [nonnegligent manslaughter](#), then calculated rates per 100,000. The population data is from the [U.S. Census Bureau](#). Murder rates were calculated based on the [FBI Uniform Crime Reports](#) and the estimated 2015 population of each state.

The 2015 U.S. population total was 320.9 million. The 2015 U.S. overall murder rate per 100,000 inhabitants was 4.89.

States

State	Population (total inhabitants) (2015) ^[1]	Murders and Nonnegligent Manslaughter (total deaths) (2015) ^[2]	Murder and Nonnegligent Manslaughter Rate (per 100,000 inhabitants) (2015)
Alabama	4,853,875	348	7.2
Alaska	737,709	59	8.0
Arizona	6,817,565	309	4.5
Arkansas	2,977,853	181	6.1
California	38,993,940	1,861	4.8
Colorado	5,448,819	176	3.2
Connecticut	3,584,730	117	3.3
Delaware	944,076	63	6.7
District of Columbia	670,377	162	24.2

https://en.wikipedia.org/w/index.php?title=Gun_violence_in_the_United_States_by_state&direction=prev&oldid=810166167

- 
- Unfortunately, there is no link to a data file. To access the data table in R, we had to do some web scraping.
 - Web scraping, or web crawling, is the term we use to describe the process of extracting data from a website.
 - Information used to render a webpage is formatted in hyper text markup language (HTML). In Chrome, you can use **Ctrl+U**/**Cmd+option+U** to access:

```
Line wrap ☐
1 <!DOCTYPE html>
2 <html class="client-nojs vector-feature-language-in-header-enabled vector-feature-language-in-main-page-header-disab
3 <head>
4 <meta charset="UTF-8">
5 <title>Gun death and violence in the United States by state - Wikipedia</title>
6 <script>(function(){var className="client-js vector-feature-language-in-header-enabled vector-feature-language-in-ma
7 RLSTATE={"ext.globalCssJs.user.styles":"ready","site.styles":"ready","user.styles":"ready","ext.globalCssJs.user":"r
8 <script>(RLQ=window.RLQ||[]).push(function(){mw.loader.impl(function(){return["user.options@12s5i",function($,jQuery
9 }]);});});</script>
10 <link rel="stylesheet" href="/w/load.php?lang=en&modules=ext.cite.styles%7Cext.uls.interlanguage%7Cext.visualEdi
11 <script async="" src="/w/load.php?lang=en&modules=startup&only=scripts&raw=1&skin=vector-2022"></scr
12 <meta name="ResourceLoaderDynamicStyles" content="">
13 <link rel="stylesheet" href="/w/load.php?lang=en&modules=site.styles&only=styles&skin=vector-2022">
14 <meta name="generator" content="MediaWiki 1.45.0-wmf.15">
15 <meta name="referrer" content="origin">
16 <meta name="referrer" content="origin-when-cross-origin">
17 <meta name="robots" content="noindex,nofollow,max-image-preview:standard">
18 <meta name="format-detection" content="telephone=no">
19 <meta property="og:image" content="https://upload.wikimedia.org/wikipedia/commons/thumb/5/50/1999-_Gun-related_death
20 <meta property="og:image:width" content="1200">
21 <meta property="og:image:height" content="900">
22 <meta property="og:image" content="https://upload.wikimedia.org/wikipedia/commons/thumb/5/50/1999-_Gun-related_death
23 <meta property="og:image:width" content="800">
24 <meta property="og:image:height" content="600">
25 <meta property="og:image:width" content="640">
26 <meta property="og:image:height" content="480">
27 <meta name="viewport" content="width=1120">
28 <meta property="og:title" content="Gun death and violence in the United States by state - Wikipedia">
29 <meta property="og:type" content="website">
30 <link rel="preconnect" href="//upload.wikimedia.org">
31 <link rel="alternate" media="only screen and (max-width: 640px)" href="//en.m.wikipedia.org/wiki/Gun_death_and_viole
32 <link rel="alternate" type="application/x-wiki" title="Edit this page" href="/w/index.php?title=Gun_death_and_violen
33 <link rel="apple-touch-icon" href="/static/apple-touch/wikipedia.png">
```

Parse HTML in R

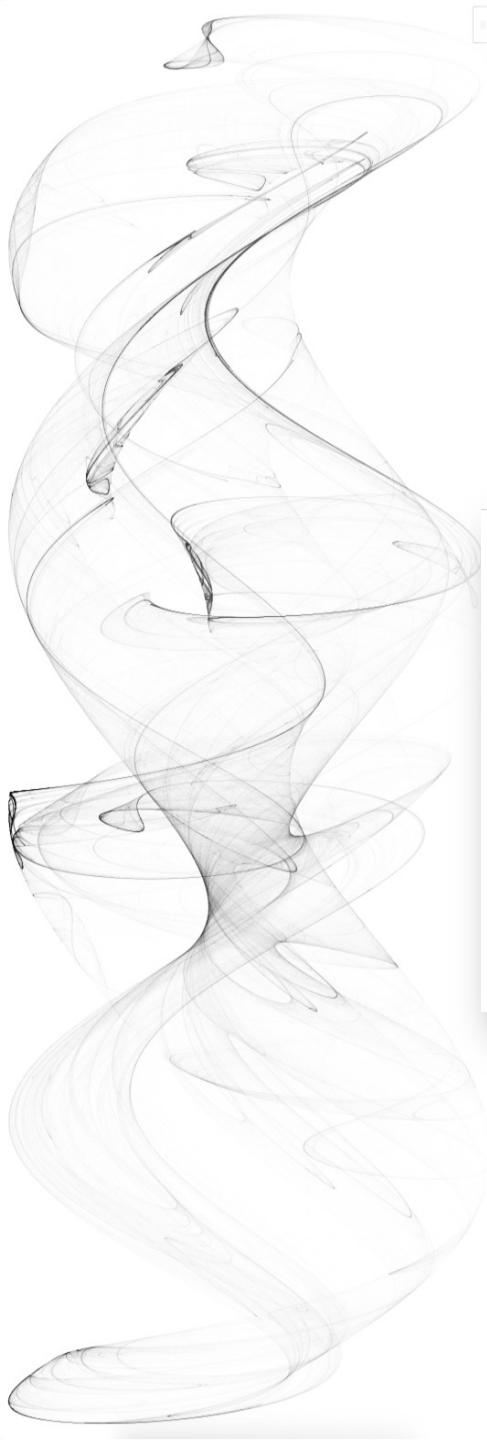
An HTML page consists of a `<html>` root element, with `<head>` for metadata (e.g., title, styles, scripts) and `<body>` for visible content (e.g., headings, paragraphs, images, links, lists, and forms).

HTML uses various elements to structure and display content. Here's an example from the Wikipedia page:

You can actually see the data wrapped by html element tags (e.g., `<td>`). We can also see a pattern of how it is stored.

If you know HTML, you can write programs that leverage knowledge of these patterns to extract what we want

```
587 <table class="wikitable sortable">
588 <tbody><tr>
589 <th>State
590 </th>
591 <th><a href="/wiki/List of U.S. states and territories by
592 </th>
593 <th>Murders and Nonnegligent
594 <p>Manslaughter<br /> <small>(total deaths)</small> <br /
595 </p>
596 </th>
597 <th>Murder and Nonnegligent
598 <p>Manslaughter Rate<br /> <small>(per 100,000 inhabitant
599 </p>
600 </th></tr>
601 <tr>
602 <td><a href="/wiki/Alabama" title="Alabama">Alabama</a>
603 </td>
604 <td>4,853,875
605 </td>
606 <td>348
607 </td>
608 <td>7.2
609 </td></tr>
610 <tr>
611 <td><a href="/wiki/Alaska" title="Alaska">Alaska</a>
612 </td>
613 <td>737,709
614 </td>
615 <td>59
616 </td>
617 <td>8.0
618 </td></tr>
619 <tr>
620 <td><a href="/wiki/Arizona" title="Arizona">Arizona</a>
621 </td>
622 <td>6,817,565
623 </td>
624 <td>309
625 </td>
626 <td>4.5
627 </td></tr>
```

Parse HTML in R

tidyverse provides a web scraping package called `rvest`.

Use `read_html` to import the webpage into R:

```
> library(rvest)
> h <- read_html(url)
> h
{html_document}
<html class="client-nojs vector-feature-language-in-header-enabled vector-fea
ture-language-in-main-page-header-disabled vector-feature-language-alert-in-s
idebar-enabled vector-feature-sticky-header-disabled vector-feature-page-tool
s-disabled vector-feature-page-tools-pinned-disabled vector-feature-main-menu
-pinned-disabled vector-feature-limited-width-enabled vector-feature-limited-
width-content-enabled" lang="en" dir="ltr">
[1] <head>\n<meta http-equiv="Content-Type" content="text/html; chars ...
[2] <body class="skin-vector skin-vector-search-vue vector-toc-pinned ...
```

Ethics and legalities of web scraping

- Ethical & legal guidelines of web scraping
 - Respect `robots.txt`; Follow website terms of service;
 - Rate limit your requests (avoid overloading servers);
 - Only scrape public data;

polite

build passing codecov 25% CRAN 0.1.3 lifecycle maturing R-CMD-check no status

The goal of `polite` is to promote responsible web etiquette.

“bow and scrape” (verb):

1. To make a deep bow with the right leg drawn back (thus scraping the floor), left hand pressed across the abdomen, right arm held aside.
2. (*idiomatic, by extension*) To behave in a servile, obsequious, or excessively polite manner. [1]
Source: *Wiktionary, The free dictionary*

The package’s two main functions `bow` and `scrape` define and realize a web harvesting session. `bow` is used to introduce the client to the host and ask for permission to scrape (by inquiring against the host’s `robots.txt` file), while `scrape` is the main function for retrieving data from the remote server. Once the connection is established, there’s no need to bow again. Rather, in order to adjust a scraping URL the user can simply nod to the new path, which updates the session’s URL, making sure that the new location can be negotiated against `robots.txt`.

The three pillars of a `polite` session are **seeking permission, taking slowly and never asking twice**.



To learn more: <https://cran.r-project.org/web/packages/polite/readme/README.html>

This abstract graphic design features a complex, swirling, and layered pattern in shades of gray and white. The composition is highly dynamic, with numerous overlapping, curved lines that create a sense of movement and depth. The overall shape is organic and fluid, resembling a stylized, flowing figure or a dynamic, organic shape. The use of varying line weights and transparency adds to the intricate and ethereal quality of the artwork.

We can display the raw code of the downloaded webpage using `html_text`, which looks messy.

Use `html_elements` to extract contents of interest. Here we are extracting tables. *Which element(s) shall we use?*

```
> html_text(h)
[1] "Gun violence in the United States by state - Wikipediadocument.documentE
lement.className=\"client-js vector-feature-language-in-header-enabled vector
-feature-language-in-main-page-header-disabled vector-feature-language-alert-
in-sidebar-enabled vector-feature-sticky-header-disabled vector-feature-page-
tools-disabled vector-feature-page-tools-pinned-disabled vector-feature-main-
menu-pinned-disabled vector-feature-limited-width-enabled vector-feature-limi
ted-width-content-enabled\";(function(){var cookie=document.cookie.match(/(?:
^|; )enwikimwclientprefs=([&]+)/);if(cookie){var featureName=cookie[1];docum
ent.documentElement.className=document.documentElement.className.replace feat
ureName+'-enabled',featureName+'-disabled'}}());RLCONF={\"wgBreakFrames\":f
alse,\"wgSeparatorTransformTable\":[\"\\\",\\\"\\\"],\"wgDigitTransformTable\":
[\"\\\",\\\"\\\"],\"wgDefaultDateFormat\":\"dmy\", \"wgMonthNames\":[\"\\\", \"January
\\\", \"February\\\", \"March\\\", \"April\\\", \"May\\\", \"June\\\", \"July\\\", \"August\\\", \"Se
```

```
> tab <- h |> html_elements("table")
> tab
{xml_nodeset (2)}
[1] <table class="wikitable sortable"><tbody>\n<tr>\n<th>State\n</th>\n<th>\n<a href="#">
[2] <table class="nowraplinks hlist mw-collapsible mw-collapsed navbox-inner" sty
```

Parse HTML in R

```
> tab[[1]] |> html_table()
# A tibble: 51 x 4
  State      Population (total inhabitants)...1 Murde...2 Murde...3
  <chr>      <chr>      <chr>      <dbl>
1 Alabama    4,853,875      348        7.2
2 Alaska     737,709       59         8
3 Arizona    6,817,565     309        4.5
4 Arkansas   2,977,853     181        6.1
5 California 38,993,940    1,861       4.8
6 Colorado   5,448,819     176        3.2
7 Connecticut 3,584,730     117        3.3
8 Delaware   944,076       63         6.7
9 District of Columbia 670,377     162       24.2
10 Florida    20,244,914    1,041       5.1
```

```
> tab[[2]] |> html_table()
# A tibble: 11 x 2
  .mw-parser-output .navbar{display:inline;font-size:88%;font...1 .mw-p...2
  <chr>      <chr>
1 "List of states and territories of the United States" "List ..."
2 "Demographics" "Popul..."
3 "Economy" "Billi..."
4 "Environment" "Botan..."
5 "Geography" "Area\..."
6 "Government" "Agric..."
7 "Health" "Ferti..."
8 "History" "Date ..."
9 "Law" "Abort..."
10 "Miscellaneous" "Abbre..."
11 "Category\nCommons\n Portals" "Catea..."
```

Use `html_table` to convert the extracted nodes to R data frames, see which one is the one you need

Parse HTML in R

```
> murders <- tab[[1]] |>  
+   html_table() |>  
+   setNames(c("state", "population", "total", "murder_rate"))  
>  
> murders |> head(n = 3)  
# A tibble: 3 × 4  
  state      population total murder_rate  
  <chr>      <chr>      <chr>      <dbl>  
1 Alabama  4,853,875    348         7.2  
2 Alaska   737,709      59          8  
3 Arizona  6,817,565    309         4.5
```

Convert, set column names, and store

We will learn how to clean up the contents (e.g., commas in the numbers) in the following lectures.

Parse HTML in R

Find elements for
contents of interest using
SelectorGadget



SelectorGadget

<http://selectorgadget.com/> 4.5 ★ (112 ratings) [Share](#)

Extension

Developer Tools

100,000 users

To learn more:

<https://rvest.tidyverse.org/articles/selectorgadget.html>

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Chapter 23 Web scraping

The data we need to answer a question is not always in a spreadsheet ready for us to read. For example, the US murders dataset we used in the R Basics chapter originally comes from this Wikipedia page:

```
url <- paste0("https://en.wikipedia.org/w/index.php?title=",  
              "Gun violence in the United States by state",  
              "&direction=prev&oldid=810166167")
```

div pre

You can see the data table when you visit the webpage:

Gun violence in the United States by state

This article is a list of the U.S. states and the District of Columbia, with population, murder and nonnegligent manslaughter, then calculated rates per 100,000. The population data is from the U.S. Census Bureau. Murder rates were calculated based on the FBI Uniform Crime Reports and the estimated 2015 population of each state.

The 2015 U.S. population total was 381.9 million. The 2015 U.S. overall murder rate per 100,000 inhabitants was 6.65.

State	Population (2015)	Murders and Nonnegligent Manslaughter (2015)	Murder and Nonnegligent Manslaughter Rate (per 100,000 inhabitants) (2015)
Alabama	4,803,075	148	7.2
Alaska	731,726	59	8.0
Arizona	6,417,100	349	4.8
Arkansas	2,977,023	181	6.1
California	38,903,000	1,481	4.8

(Web page courtesy of Wikipedia⁷⁶. CC-BY-SA-3.0 license⁷⁷. Screenshot of part of the page.)

Unfortunately, there is no link to a data file. To make the data frame that is

.r

Clear (30)

Toggle Position

XPath

?

Micro

Web scraping, or web harvesting, is the term we use to describe the